

# DataMind: Artifact and Evaluation Protocol

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## 1 Artifact and Evaluation Protocol

The artifact package will contain the DataMind prototype, the benchmark adapter, build manifests, task-level traces, and scripts for reproducing every reported aggregate. All model runs record the model identifier, prompt template, tool schema, profile, snapshot, seed, latency, and token usage.

### 1.1 Benchmark adapter checks

Before running a language model, the adapter checks that every WorkSurface-Bench table task reproduces its executable DuckDB result, every document task maps to the annotated source span, and every graph task maps to the annotated source edge. Gold surface labels are used only by the oracle settings and never by the normal DataMind serving path.

### 1.2 Build and update traces

Each build emits a manifest containing source checksums, parser and embedder versions, surface revisions, validation outcomes, and the publication event. Update traces contain the previous snapshot, changed sources, generated revisions, and the first snapshot in which the update is visible. Failed builds retain completed stage receipts so that recovery can resume without silently reprocessing published artifacts.

### 1.3 Policy and recovery checks

The artifact includes deterministic tests for profile isolation, path allowlisting, read-only SQL, destructive-operation confirmation, and audit receipt integrity. Fault-injection scripts disable or corrupt one surface at a time and record fallback decisions, evidence receipts, recovery latency, and final answer correctness.

### 1.4 Local reproduction

```
git clone https://github.com/OpenDCAI/DataMind.git
cd DataMind
python -m venv .venv && source .venv/bin/activate
pip install -e .
python -m datamind.scripts.seed_enterprise_demo
python -m pytest datamind/tests
```

The MCP adapter is located under <https://github.com/OpenDCAI/DataMind/tree/main/plugins/datamind-context>. It is a deployment adapter over the same profile data plane and is not a separate evaluation system.

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*SIGMOD '27, Huntington Beach, CA, USA*

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ACM ISBN 979-8-4007-XXXX-X/27/06

<https://doi.org/10.1145/XXXXXXX.XXXXXXX>

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